

MEMORY IN MOTION: PREDICTIVE 4D BELIEF FOR NAVIGATION IN EVOLVING WORLDS

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ABSTRACT

Object navigation usually assumes that a target remains where it was mapped, yet a long-lived household agent often acts on observations that are hours or days old. Portable objects may move while unobserved, making the last-seen location an unreliable navigation goal. We study *persistent dynamic object navigation*: an agent must locate an object from a partial, temporally stale history, gather physical evidence, and revise its route when memory and observation disagree. We introduce P4D-NAV, which turns an evidence-grounded 4D memory into a predictive belief over the target’s current state. A continuous-time Transformer encodes irregular observation histories, while a structured persistence–relocation pointer head separately estimates whether the last state remains valid and where the object moved otherwise. A visibility-aware Bayesian update converts positive and negative observations into posterior beliefs, enabling the same agent to navigate directly when belief is concentrated, search candidates when it is diffuse, and replan after failed inspections. We further introduce P4D-BENCH, a suite of paired HSSD navigation episodes whose hidden changes are compiled from complementary human activity, object-interaction, and household-state records. The benchmark controls scene, target, start pose, and candidate set across static, routine, and random worlds, isolating whether navigation gains arise from learnable dynamics rather than frequency shortcuts. Experiments evaluate predictive navigation, belief-guided search, evidence-aware recovery, online changes, and cross-scene generalization; numerical results are intentionally left blank in this working draft pending completion of the navigation rollouts.

1 INTRODUCTION

Object navigation asks an embodied agent to find an instance of a requested category from egocentric observations. Progress in semantic mapping and foundation-model reasoning has made this problem increasingly tractable in static environments (Chaplot et al., 2020; Huang et al., 2023; Gu et al., 2024). A household robot deployed over weeks, however, rarely acts in the scene that it originally mapped. A mug last observed beside the sink may have moved to a desk after breakfast; keys may alternate between a shelf and a bag; and a rarely moved appliance may remain exactly where it was. In each case, the agent has a valid historical observation but an uncertain current navigation goal.

This distinction exposes a limitation of conventional embodied memory. Most spatial memories answer *where was the object observed?*; navigation requires *where is it likely to be now?* More importantly, a predictive answer should not be consumed as a static waypoint. Travelling takes time, candidate locations differ in inspection cost and visibility, and a failed inspection is new evidence. A useful memory must therefore support a closed loop: retrieve relevant history, propagate it to the query time, select a cost-effective location, observe, update the belief, and replan.

Several recent lines of work approach parts of this problem. Dynamic scene memories update object states after new observations (Liu et al., 2024; Kurenkov et al., 2023); predictive scene graphs estimate the persistence or re-emergence of time-varying relations (Saavedra-Ruiz et al., 2026); and models of household routines predict multimodal future object placements for downstream navigation (Patel & Chernova, 2023; Argenziano et al., 2026). Portable Object Navigation goes further by allowing a target to move while the agent travels and learning transit-aware policies (Dorbala et al., 2026). These methods establish that dynamics matter, but leave a complementary question open: how should an

agent turn its own long-term, partially observed memory into a belief that directly controls physical search and is corrected by the evidence collected along the route?

We formulate this setting as **persistent dynamic object navigation**. Before receiving a query, the agent has accumulated a timestamped 4D memory through intermittent visits. The target may undergo one or more hidden transitions after its last positive observation. At query time, the agent receives neither the current state nor the activity that caused the transition. It must find the object under a navigation budget using only causal memory, a map of candidate receptacles and viewpoints, and online RGB-D observations. Our main protocol freezes the target once the query begins to isolate the value of predictive memory; an extension allows an additional hidden transition during navigation.

We introduce P4D-NAV, a compact predictive-memory agent for this problem. Its 4D memory maintains persistent entity identities, time-valid state versions, and observation provenance in a common 3D frame. A target-conditioned retriever exposes only the history available before the query. A lightweight continuous-time Transformer represents irregular temporal gaps and calendar periodicity. Instead of directly classifying a location, a structured head factorizes the belief into (i) the probability that the last-seen state persists and (ii) a pointer distribution over relocation candidates. This inductive bias matches the physical alternatives faced by a navigator and supports candidate sets that vary across scenes.

Prediction is coupled to action through a visibility-aware evidence update and a belief-guided controller. Concentrated beliefs support a direct first attempt; diffuse beliefs induce a probability–cost search order; and sufficiently covered negative observations suppress inspected hypotheses before replanning. The controller is identical across stable, routine-mobile, personal-habit, and irregular objects and receives no mobility label at test time. Thus, static recall, predictive navigation, and uncertainty-driven search emerge from one belief rather than from an oracle router.

To evaluate the resulting navigation problem, we introduce P4D-BENCH. Its dynamic HSSD episodes are compiled through a canonical routine representation that combines complementary signals from longitudinal activity records, real human–object interactions, and household object-state scripts. More importantly, the benchmark contains paired static, routine, and random worlds that share the scene, target, last observation, query time, candidate set, and agent start. Only the hidden transition process differs. This counterfactual construction tests whether an agent exploits temporal structure instead of object–location frequency or route-length shortcuts.

Contributions. Our contributions are threefold:

- We formulate persistent dynamic object navigation, connecting long-term, partially observed memory to closed-loop physical search under hidden object transitions.
- We propose P4D-NAV, which combines an evidence-grounded 4D memory, continuous-time history encoding, a persistence–relocation belief, and visibility-aware Bayesian correction to support direct navigation, belief-guided search, and recovery without oracle dynamics labels.
- We introduce P4D-BENCH, a human-trace-grounded HSSD benchmark with paired dynamic controls, instance-level mobility regimes, online replanning tasks, and cross-scene and cross-household evaluation.

2 RELATED WORK

Object navigation and active search. ObjectNav benchmarks couple semantic perception, exploration, and navigation to an unseen target (Anderson et al., 2018; Savva et al., 2019; Chaplot et al., 2020). Semantic maps and open-vocabulary scene representations improve generalization by associating language with metric or graph-structured space (Huang et al., 2023; Gu et al., 2024; Werby et al., 2024). Probabilistic object-search methods use object–room or object–object correlations to order exploration and update a belief after observations (Zheng et al., 2022; Ginting et al., 2024). These methods primarily search for a target whose state is fixed within an episode. P4D-NAV addresses a different source of uncertainty: the agent has previously observed the target, but that observation may have become stale because the environment evolved between visits.

Navigation in changing environments. Early work on dynamic mapping often filters moving content or updates a semantic map after changes are observed (Bescos et al., 2018; Narayana et al., 2020). Scene Graph Memory models a partially observable dynamic graph and predicts object–location links (Kurenkov et al., 2023), while DynaMem continually reconstructs an actionable spatio-semantic memory for mobile manipulation (Liu et al., 2024). Predictive spatio-temporal scene graphs attach persistence and emergence models to open-vocabulary relations (Saavedra-Ruiz et al., 2026). FlowMaps learns multimodal continuous future object placements and uses ranked hypotheses for dynamic ObjectNav (Argenziano et al., 2026). Portable Object Navigation and Transit-Aware Planning model objects that move as the agent moves (Dorbala et al., 2026). Our setting is complementary: dynamics are inferred from the agent’s causal, intermittent history; object-specific regimes are hidden; and prediction is embedded in an observe–update–replan loop rather than consumed only as a fixed candidate ranking or known transition path.

Long-term embodied memory. Long-horizon benchmarks increasingly require agents to retain observations across exploration or interaction (Majumdar et al., 2024; Yadav et al., 2025; Zhang et al., 2026). Open-vocabulary 3D memories support spatial retrieval (Gu et al., 2024; Werby et al., 2024), while recent 4D representations index time-varying states or human-centered events (Sohn et al., 2025; Bartoli et al., 2026). These systems motivate persistent identity, temporal validity, and provenance. We use these properties as infrastructure, but focus the learning problem on a navigation-relevant belief: whether an entity’s last state persists and, if not, which reachable state should be inspected next.

Human routines and personalized object dynamics. Household activities exhibit temporal and person-specific regularities. HOMER and SLaTe-PRO predict routine object usage for proactive assistance (Patel & Chernova, 2023); STREAK studies continual adaptation under household context drift (Bartoli et al., 2024); and UcON evaluates navigation with a large user-habit knowledge base (Wang et al., 2026). Existing dynamic-world generators commonly rely on hand-designed routines or language-model priors. P4D-BENCH instead normalizes complementary longitudinal activity, real object-interaction, and complete state-transition records into executable HSSD episodes. Its paired worlds preserve marginal scene and query factors while changing the transition mechanism, enabling controlled measurement of when routine learning helps navigation.

3 PREDICTIVE 4D BELIEF NAVIGATION

3.1 PERSISTENT DYNAMIC OBJECT NAVIGATION

Let an environment contain a navigable space $\mathcal{X}$, a set of persistent entities $\mathcal{O}$, and candidate object states $\mathcal{C}_o$ for each target $o \in \mathcal{O}$. A state identifies a semantically valid receptacle or region and its associated inspection viewpoints. Before a query at time t_q , the agent has collected only a partial history

$$\mathcal{H}_{\leq t_q} = \{(I_k, D_k, \xi_k, t_k) \mid t_k \leq t_q\}, \quad (1)$$

where I_k, D_k, ξ_k denote RGB, depth, and camera pose. The target’s true query-time state $s_q^o \in \mathcal{C}_o \cup \{\text{unknown}\}$ and any unobserved transition events are hidden.

An episode supplies a target description, query time, initial agent pose x_0 , a causal memory materialized from $\mathcal{H}_{\leq t_q}$, and a finite action budget. At decision step j , the agent chooses a candidate viewpoint or a low-level navigation action, receives an observation z_j , and may terminate with STOP or NOT_FOUND. Success requires a correct stop near a valid goal viewpoint with the target visible. The policy should maximize success while minimizing travelled distance and inspections:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi} [\mathbb{I}(\text{success}) - \lambda_d L - \lambda_n N_{\text{inspect}}]. \quad (2)$$

The main protocol holds the target fixed after t_q , isolating whether a stale history can be transformed into a useful current belief. The online-dynamic extension advances the world clock during execution and may apply an additional transition while the target is outside the camera view.

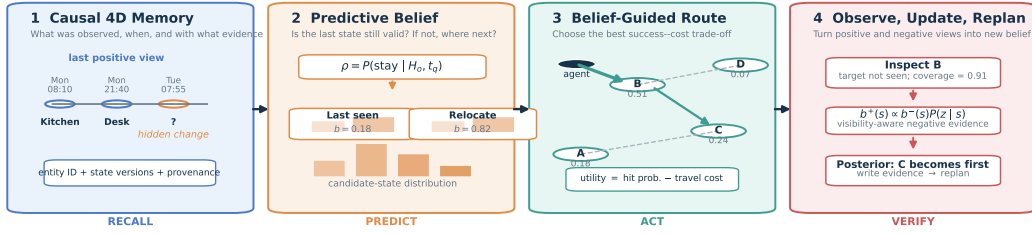


Figure 1: Overview of P4D-NAV. A causal 4D memory is propagated to the query time as a structured persistence–relocation belief. The agent trades target probability against travel cost, converts visibility-qualified observations into posterior evidence, and replans after unsuccessful inspections.

3.2 EVIDENCE-GROUNDED 4D MEMORY

The 4D memory represents a persistent 3D semantic scene whose entity states evolve over time. For image coordinate $\tilde{\mathbf{u}} = [u, v, 1]^\top$ with depth $D_t(u, v)$, the corresponding world point is

$$\mathbf{x}^w = \text{proj}_3 \left(\mathbf{T}_{w \leftarrow c, t} \begin{bmatrix} D_t(u, v) \mathbf{K}^{-1} \tilde{\mathbf{u}} \\ 1 \end{bmatrix} \right), \quad (3)$$

where $\mathbf{K}$ is the camera intrinsic matrix and $\mathbf{T}_{w \leftarrow c, t}$ is the camera-to-world transform. Cross-view association consolidates observations using semantic compatibility, 3D overlap, temporal consistency, and source confidence. Repeated observations strengthen one persistent entity rather than creating disconnected detections.

Memory is materialized causally from versioned deltas:

$$\mathcal{M}_{\leq t} = \text{Materialize}(\Delta \mathcal{M}_0, \dots, \Delta \mathcal{M}_t) = (\mathcal{G}_t, \mathcal{V}_t, \mathcal{B}_t), \quad (4)$$

where $\mathcal{G}_t$ stores semantic and spatial relations, $\mathcal{V}_t$ stores time-valid entity versions, and $\mathcal{B}_t$ stores observation provenance. The m -th version of entity o is

$$h_o^m = ([t_{o,s}^m, t_{o,e}^m], s_o^m, c_o^m, f_o^m, \mathcal{P}_o^m), \quad (5)$$

containing a validity interval, candidate state, confidence, visual feature, and evidence handles. Persistent identity is separated from mutable state: moving an object appends a version without erasing its earlier trajectory. All query-time views exclude observations and events after t_q .

3.3 TARGET-CONDITIONED HISTORY RETRIEVAL

For a target o , the retriever first resolves its persistent identity or category-compatible instances, then returns the most recent K causal events

$$H_o = \{e_k = (o_k, s_k, t_k, z_k, c_k, v_k)\}_{k=1}^K. \quad (6)$$

Here s_k is the observed candidate state; $z_k \in \{\text{seen}, \text{notseen}\}$ is the evidence type; c_k is perception confidence; and v_k summarizes whether the hypothesized state was adequately visible, including frustum coverage and unoccluded fraction. Context events may be included when they share an entity, receptacle, region, or time window with the target, but hidden activity labels are never exposed at test time.

Candidates $i \in \mathcal{C}_o$ are not global class indices. Each is encoded from its region, receptacle category, functional role, geometry, and instance identity. This pointer-compatible representation permits the number and identity of states to vary across homes and supports evaluation on unseen receptacle instances. An additional `unknown` candidate absorbs states outside the mapped set.

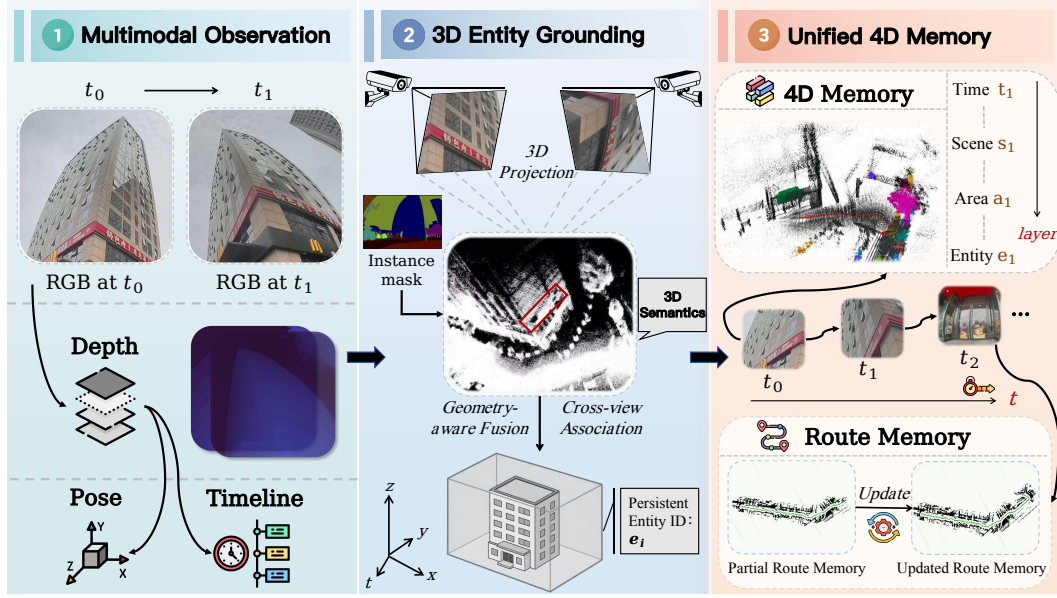


Figure 2: 4D memory construction. Multimodal observations are registered in a shared frame, consolidated into persistent entities, and retained as temporally versioned states with supporting evidence.

3.4 CONTINUOUS-TIME HISTORY ENCODER

Each event token combines entity, state, evidence, confidence, and visibility embeddings with an explicit encoding of elapsed and calendar time:

$$\mathbf{x}_k = E_o(o_k) + E_s(s_k) + E_z(z_k) + E_c(c_k) + E_v(v_k) \quad (7)$$

$$+ \phi_\Delta(t_q - t_k) + \phi_{\text{cal}}(t_k), \quad (8)$$

where ϕ_Δ includes $\log(1 + t_q - t_k)$ and ϕ_{cal} contains sinusoidal hour-of-day and day-of-week features. A learned query token specifies the target, query time, and last positively observed state. A small Transformer produces contextual event features and query representation

$$(\mathbf{h}_1, \dots, \mathbf{h}_K, \mathbf{h}_q) = \text{CTTransformer}(\mathbf{x}_1, \dots, \mathbf{x}_K, \mathbf{x}_q). \quad (9)$$

Unlike a fixed time-bin transition table, this encoder conditions on irregular gaps, repeated negative evidence, and the complete order of observed states. It does not receive oracle mobility type or the activity that generated a hidden transition.

3.5 PERSISTENCE-RELOCATION BELIEF

A direct classifier entangles two physically distinct questions: whether anything changed and where the object went. We instead factorize them. Let s_{last} be the last positive state. The persistence branch predicts

$$\rho_q = P(s_q^o = s_{\text{last}} \mid H_o, t_q) = \sigma(\mathbf{w}_\rho^\top \mathbf{h}_q). \quad (10)$$

For every relocation candidate $i \neq s_{\text{last}}$, a shared pointer head computes

$$a_i = \frac{(W_q \mathbf{h}_q)^\top (W_c \mathbf{g}_i)}{\sqrt{d}} + \psi(\mathbf{h}_q, \mathbf{g}_i), \quad \pi_q(i) = \frac{\exp a_i}{\sum_{j \neq s_{\text{last}}} \exp a_j}, \quad (11)$$

where $\mathbf{g}_i$ is the candidate representation and ψ captures target-candidate compatibility. The prior belief is

$$b_q^-(i) = \begin{cases} \rho_q, & i = s_{\text{last}}, \\ (1 - \rho_q)\pi_q(i), & i \neq s_{\text{last}}. \end{cases} \quad (12)$$

The model is trained end-to-end with a single negative log-likelihood, $\mathcal{L}_{\text{state}} = -\log b_q^-(s_q^o)$. The factorization is therefore an inductive bias, not a separately supervised oracle switch. In static

environments it can place mass on persistence; for recurring relocations it can concentrate π_q ; and for irregular objects it can retain a broad distribution.

3.6 VISIBILITY-AWARE BAYESIAN EVIDENCE UPDATE

After navigating to candidate r , the agent obtains detection result z , detector reliability c , and effective visibility $v \in [0, 1]$. A positive instance match assigns high likelihood to r . A negative observation should suppress r only if the target would probably have been detected there. We use

$$P(z = \text{notseen} \mid s_q^o = i, c, v) = \begin{cases} 1 - cv, & i = r, \\ 1, & i \neq r, \end{cases} \quad (13)$$

followed by

$$b_q^+(i) = \frac{b_q^-(i)P(z \mid s_q^o = i, c, v)}{\sum_j b_q^-(j)P(z \mid s_q^o = j, c, v)}. \quad (14)$$

Thus, a clear empty view provides strong negative evidence, whereas an occluded or poorly covered view makes only a small change. The update is parameter free and can be applied repeatedly. Each inspection and its posterior are written back as a provisional memory event, preserving the causal chain used by subsequent decisions.

3.7 BELIEF-GUIDED NAVIGATION AND REPLANNING

Each candidate i has one or more reachable inspection viewpoints $\mathcal{W}_i$. Let $d(x, \mathcal{W}_i)$ be the shortest geodesic distance from the current pose and $\hat{v}_i$ the expected inspection coverage. The high-level policy selects

$$i^* = \arg \max_{i \in \mathcal{C}_o \setminus \mathcal{I}} \frac{b(i)\hat{v}_i}{d(x, \mathcal{W}_i) + \lambda_{\text{inspect}}}, \quad (15)$$

where $\mathcal{I}$ contains sufficiently inspected hypotheses. The fixed low-level navigator executes the route and the evidence update in Equation (14) produces the belief for the next decision. This separation ensures that gains reflect memory and state selection rather than a different locomotion controller.

The same interface instantiates three core tasks. *Predictive Object Navigation* executes only the top-belief candidate and measures first-go success. *Belief-Guided Object Search* follows the probability-cost policy under a candidate budget. *Evidence-Aware Replanning* updates after every qualified view and measures recovery from an incorrect first prediction. For online-dynamic episodes, the model is queried at the candidate arrival time $t + \hat{\tau}_i$ rather than the current time:

$$b_{t+\hat{\tau}_i}^- = F_\theta(H_o, t + \hat{\tau}_i), \quad (16)$$

and replanning repeats after every action chunk. No explicit router chooses between “static” and “dynamic” policies; their behavior is induced by the learned belief and the evidence observed during navigation.

4 EXPERIMENTAL SETUP

4.1 P4D-BENCH

Human-trace-grounded episode generation. P4D-BENCH instantiates persistent dynamic object navigation in Habitat using HSSD scenes (Khanna et al., 2024). Rather than sampling arbitrary relocations, we first normalize complementary sources into canonical routine records. CASAS contributes longitudinal activity timing and room occupancy (Cook et al., 2013); HD-EPIC and ParaHome contribute real object manipulation and action transitions (Perrett et al., 2025; Kim et al., 2024); and our HOMER+ expansion contributes multi-day household object-state scripts and variations derived from HOMER (Patel & Chernova, 2023). A record contains actor, activity, object, action, pre- and post-state, source and destination, time, duration, recurrence, provenance, and confidence. The compiler grounds records to semantically compatible HSSD entities, enforces receptacle affordances, collision-free placement, causal state continuity, and navigable inspection viewpoints, and records whether a grounding is exact, fallback, or proxy.

Table 1: Navigation tasks in P4D-BENCH. N1–N3 form the main evaluation; N4 and N5 test online adaptation and transfer.

	Task	Decision signal	Primary measure
N1	Predictive navigation	Top-1 belief	First-Go SR
N2	Belief-guided search	Full belief + cost	Search SR / cost
N3	Evidence replanning	Posterior belief	Recovery SR
N4	Online-dynamic nav.	Arrival-time belief	Online Recovery SR
N5	Cross-scene transfer	Shared candidate roles	Generalization SR

Table 2: Final benchmark statistics. Entries are intentionally blank until the multi-scene generation and leakage audit are complete.

Split	Scenes	Days	Targets	Episodes
Train	–	–	–	–
Validation	–	–	–	–
Test	–	–	–	–
Total	–	–	–	–

Paired worlds. For each base query we construct static, routine, and random variants. They share scene, target, last positive observation, query time, legal candidate set, agent start, and navigation budget. In the static variant the target remains at its last state. In the routine variant, hidden transitions follow the compiled temporal and instance-level regularities. In the random variant, destinations are sampled from the same legal candidates while breaking the learned dependence. Pairing prevents scene layout, target identity, and route length from explaining a routine-specific gain.

Mobility types. Every target instance is assigned one analysis label using training history only: *stable*, *activity-routine mobile*, *personal-habit mobile*, or *irregular mobile*. Let

$$r_{\text{move}}(o) = \frac{N_{\text{move}}(o)}{N_{\text{days}}}, \quad G_{\text{act}}(o) = \text{Acc}(s' \mid s, a, o) - \text{Acc}(s' \mid s, o), \quad (17)$$

and $C_{\text{habit}}(o) = \max_s P(s \mid o)$. Frozen training-set thresholds on these statistics determine the label. Labels support stratified reporting but are hidden from all test-time agents. Categories are distributed across multiple mobility types so the label cannot be inferred from the object name alone.

Task protocols. N1 allows one predicted candidate and isolates whether state accuracy transfers to physical navigation. N2 permits multiple inspections under a fixed step, distance, and candidate budget. N3 additionally requires a visibility-aware posterior update after every inspection. N4 applies another hidden transition during execution only while the target is outside the current view. N5 holds out scenes, households, object instances, or receptacle instances. The test evaluator withholds current state, target pose, valid goal viewpoints, hidden events, mobility type, and future observations.

Splits and quality control. We split complete days and households rather than individual queries, and use disjoint HSSD scenes for cross-scene evaluation. Near-duplicate event groups cannot cross splits. Each episode requires a prior positive observation, a non-unknown target state, collision-free instantiation, at least one reachable goal viewpoint, an initial target distance of at least 3 meters, and no target visibility from the start pose. Counts of scenes, households, days, events, targets, candidates, and episodes will be inserted after the final generation audit (Table 2).

4.2 BASELINES AND ABLATIONS

We compare against four families under identical candidates, viewpoints, perception, path planner, and action budget. *Non-predictive* baselines include Last Seen and Nearest Candidate Search. *Statistical* baselines include Frequency Prior, first-order Markov transitions, time-conditioned transitions, and an instance-hotspot prior. *Learned* baselines include GRU Direct, Transformer Direct, Scene Graph Memory (Kurenkov et al., 2023), a PredictiveGraphs-style filter (Saavedra-Ruiz et al., 2026), and a

Table 3: Main results on routine-world N1–N3 episodes. All methods share the same perception, candidates, inspection viewpoints, and low-level navigator. Lower is better for inspections and distance.

Method	First-Go SR $\uparrow$	Search SR $\uparrow$	SPL $\uparrow$	Inspect. $\downarrow$	Distance $\downarrow$	Recovery SR $\uparrow$
Nearest Candidate	–	–	–	–	–	–
Last Seen	–	–	–	–	–	–
Frequency Prior	–	–	–	–	–	–
Markov Transition	–	–	–	–	–	–
Time-Conditioned Prior	–	–	–	–	–	–
Transformer Direct	–	–	–	–	–	–
PredictiveGraphs-style	–	–	–	–	–	–
FlowMaps-style (Argenziano et al., 2026)	–	–	–	–	–	–
P4D-Belief Prior	–	–	–	–	–	–
P4D-NAV (ours)	–	–	–	–	–	–
Oracle Current State	–	–	–	–	–	–

FlowMaps-style future-location model (Argenziano et al., 2026). Where an external method assumes a different state space, we rank the same benchmark candidates by its predicted mass to isolate the belief model. N4 additionally compares against Transit-Aware Planning under its required online target path input (Dorbala et al., 2026). Oracle Current State provides the navigation upper bound; Oracle Activity is reported only as a diagnostic and never as a fair baseline.

The principal ablations remove: continuous-time features; the persistence–relocation factorization; instance history; calendar context; negative evidence; visibility calibration; Bayesian updating; or the full belief in favor of Top-1. *P4D-Belief Prior* predicts once and searches without updating, while full P4D-NAV closes the observe–update–replan loop.

4.3 METRICS

We report state Top-1/Top- k , negative log-likelihood, Brier score, expected calibration error, true-state rank, and persistence AUROC. Navigation metrics include Success Rate (SR), SPL, path length, completion time, First-Go SR, number of inspected candidates, Recovery SR after an incorrect first prediction, and posterior rank improvement. Because a moving target makes a fixed shortest path ambiguous, N4 additionally reports excess path length relative to a trajectory-aware oracle. All aggregate results include bootstrap confidence intervals over episodes and both micro and mobility-type macro averages.

Results are stratified by world variant, mobility type, transition class (stay, same-region move, cross-region move, multi-move, and return), staleness interval, spatial difficulty, grounding quality, and generalization split. The key causal contrast is the paired routine gain over the same method in static and random variants.

Implementation and reporting details are provided in Appendices A.7 and A.8.

5 RESULTS AND ANALYSIS

This working draft defines the complete evaluation matrix but does not invent unfinished measurements. Every dash below must be replaced by a reproduced value or a clearly cited published result before submission. The final discussion should be organized around three hypotheses: predictive belief improves the first navigation decision in routine worlds; a calibrated distribution reduces search cost when Top-1 is uncertain; and visibility-aware updates recover efficiently when the prior is wrong.

5.1 DOES PREDICTIVE MEMORY IMPROVE NAVIGATION?

First-Go SR is the cleanest measure of whether a more accurate current-state belief becomes a better physical destination. Search SR and cost then test whether probability mass beyond Top-1 is useful. Recovery SR is evaluated only on episodes in which the initial candidate was incorrect, preventing easy first-attempt successes from hiding poor replanning. The final analysis will report paired episode-level differences and confidence intervals rather than comparing only aggregate means.

5.2 WHEN DOES PREDICTIVE MEMORY HELP?

Stable objects test backward compatibility: a predictive model should not damage a strong last-seen prior. Activity-routine objects test temporal transitions; personal-habit objects test long-term instance memory beyond global activity frequencies; and irregular objects test whether a broad, well-calibrated belief improves search even when Top-1 accuracy has limited headroom. The stratified result matrix is in Table 4; we will additionally plot SR and excess path length against staleness to distinguish genuine temporal reasoning from memorized hotspots.

5.3 DOES THE GAIN COME FROM LEARNABLE DYNAMICS?

The paired-world matrix is reported in Table 5. The expected signature is not uniformly high performance: P4D-NAV should preserve static performance, gain most in routine worlds, and express greater uncertainty in marginal-matched random worlds. An improvement that remains unchanged after routine transitions are randomized would indicate a frequency, category, or route-length shortcut rather than learned dynamics.

5.4 ABLATIONS, GENERALIZATION, AND FAILURE ANALYSIS

The final ablation table will jointly report state NLL and navigation cost, revealing whether a representation improves only classification or also changes useful decisions. In particular, we will compare query-time versus arrival-time belief on N4, prior-only versus Bayesian updating on N3, and structured versus direct heads on cross-scene candidates. Generalization results will separate seen-scene/unseen-household, unseen-scene/unseen-household, unseen object instance, and unseen receptacle instance.

Failures will be categorized as entity association, missing candidate, stale-state prediction, overconfident relocation, insufficient visibility, detector error, unreachable viewpoint, or replanning error. Qualitative trajectories will show the prior distribution, route, inspection coverage, posterior, and next decision at each step. This evidence is necessary to establish that improved navigation follows from the claimed closed-loop mechanism.

6 CONCLUSION

We introduced persistent dynamic object navigation, a setting in which an embodied agent must act on a long-term but potentially stale history of a changing environment. P4D-NAV treats 4D memory as a predictive belief rather than a last-seen database: it separates persistence from relocation, represents variable scene-specific candidates, converts visibility-qualified observations into posterior evidence, and uses the evolving distribution to navigate, search, and replan. P4D-BENCH grounds this question in human-trace-driven HSSD episodes and paired static, routine, and random worlds. The resulting formulation evaluates a simple but consequential capability for long-lived agents: navigating not merely to where an object was observed, but to where the accumulated evidence indicates it is now. Quantitative conclusions will be added only after the multi-scene navigation evaluation is complete.

7 LIMITATIONS

P4D-NAV assumes reliable entity association and an available set of candidate receptacles and inspection viewpoints; pose drift, missed detections, identity switches, or missing candidates can corrupt both prediction and navigation. Its discrete belief also represents arbitrary continuous placements less directly than continuous generative models. The source datasets are complementary rather than jointly observed, so p4dbench episodes are human-trace-grounded simulations, not reconstructions of one household. The main protocol freezes the target after the query; N4 is required to test simultaneous target and agent motion. Finally, visibility-aware updates reduce weak negative evidence but cannot guarantee recovery when beliefs and perception fail together or routines shift.

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A SUPPLEMENTARY MATERIAL

A.1 FORMAL EPISODE INTERFACE

Each navigation episode references one causal prediction query. Public fields include schema version, episode and query identifiers, split, task type, scene and household identifiers, world variant, target description, query time, agent start pose, candidate-state identifiers, budgets, and success criteria. Evaluation-private fields include the current state, target pose, valid goal viewpoints, hidden transition class, oracle path, mobility label, and query-time activity. Train and validation releases may expose analysis labels in metadata; the formal test release replaces them with null values and computes grouped metrics in a private evaluator.

The high-level track exposes

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NAVIGATE_TO(state), INSPECT(state), STOP, NOT_FOUND,
```

with a fixed navigation backend. The low-level track exposes Habitat-style forward, turn, look, and stop actions. The main paper uses the high-level track to isolate predictive state selection; low-level results measure sensitivity to control and perception.

A.2 EPISODE VALIDITY AND LEAKAGE AUDIT

An episode is retained only if: (i) the target has at least one positive pre-query observation; (ii) the target state is known to the private evaluator and physically instantiated without collision; (iii) at least one valid viewpoint lies on the NavMesh; (iv) the start is on the NavMesh, at least 3 meters from the target, and does not reveal it; and (v) all public histories terminate at or before the query. A successful stop must be within 1 meter geodesic distance of a valid viewpoint, observe the target above a frozen visible-fraction threshold, and identify the correct instance or category.

We audit serialized examples for future timestamps, hidden event fields, filenames or indices correlated with answers, duplicate event groups across splits, and inconsistent paired variants. Static, routine, and random members of a pair must have identical public query fields. Invalid simulator episodes are reported separately and are not counted as ordinary failures.

A.3 MOBILITY-TYPE ASSIGNMENT

The frozen rule is

$$\text{type}(o) = \begin{cases} \text{stable}, & r_{\text{move}}(o) < \tau_{\text{move}}, \\ \text{activity}, & G_{\text{act}}(o) \geq \tau_{\text{act}}, \\ \text{personal}, & C_{\text{habit}}(o) \geq \tau_{\text{habit}}, \\ \text{irregular}, & \text{otherwise.} \end{cases} \quad (18)$$

All thresholds are selected on training data before navigation evaluation. When the generator specifies a mobility profile, the same statistics are used to verify that realized trajectories exhibit the intended behavior. Every group must cover several categories, instances, and scenes, and no category may uniquely reveal one mobility type.

A.4 METRIC DEFINITIONS

Let S_i be episode success, L_i the executed path, and L_i^* the oracle shortest path to the query-time goal under the same start. Standard SPL is

$$\text{SPL} = \frac{1}{N} \sum_{i=1}^N S_i \frac{L_i^*}{\max(L_i, L_i^*)}. \quad (19)$$

First-Go SR counts success when only the initial predicted candidate may be inspected. Recovery SR conditions on an incorrect first candidate and measures whether the target is subsequently found. Excess path is $L_i - L_i^*$ for static-after-query episodes. For online-dynamic episodes, the oracle is trajectory aware and may intercept the target; we therefore report both success within the common budget and $L_i - L_{i,\text{dyn}}^*$.

Calibration metrics operate on the full candidate belief before navigation and after each update. ECE bins maximum confidence, Brier score evaluates all candidates, and true-state rank avoids hiding useful changes below Top-1. Search-cost curves plot success against allowed inspections and travelled distance rather than reporting one arbitrary budget.

A.5 FULL BASELINE SPECIFICATION

Last Seen assigns all mass to the latest positive state. Frequency Prior estimates $P(s \mid o)$ from training data. Markov Transition estimates $P(s_{k+1} \mid s_k, o)$; Time-Conditioned Prior additionally conditions on public hour and weekday bins. Instance Hotspot uses only the target instance’s training trajectory. GRU Direct and Transformer Direct receive the same event tokens and candidate encoder as P4D-NAV but directly normalize candidate logits without the persistence factorization. P4D-Belief Prior uses Equation (12) once at episode start. Full P4D-NAV applies Equation (14) and Equation (15) after each inspection.

The Oracle Activity diagnostic may use the hidden activity and source location but is excluded from fair rankings. Oracle Current State receives the private target state and measures remaining perception and navigation error. External predictive methods are adapted only through their public inputs; any use of privileged simulator state must be identified and excluded from the main comparison.

A.6 DISAGGREGATED RESULT TEMPLATES

Table 4: Search SR by hidden mobility type. Types are determined from training history and withheld from the agent. Macro average weights the four groups equally.

Method	Stable	Activity routine	Personal habit	Irregular	Macro
Last Seen	–	–	–	–	–
Time-Conditioned	–	–	–	–	–
Instance Hotspot	–	–	–	–	–
Transformer Direct	–	–	–	–	–
P4D-NAV	–	–	–	–	–

Table 5: Paired-world navigation results. Variants share all observed setup variables and differ only in hidden transitions.

Method	Static SR	Routine SR	Random SR	Routine gain over Last Seen
Last Seen	–	–	–	–
Markov Transition	–	–	–	–
Transformer Direct	–	–	–	–
P4D-NAV	–	–	–	–

A.7 IMPLEMENTATION DETAILS

Unless otherwise noted, the history encoder has three Transformer layers, hidden width 128, four attention heads, and at most 64 events. The predictive model contains approximately –million trainable parameters and is optimized with AdamW using validation NLL for checkpoint selection. Candidate encoders are shared across scenes. The main high-level track fixes candidate viewpoints and low-level planning; a Habitat low-level action track evaluates the complete embodied stack. Final learning rate, batch size, epoch count, hardware, simulator version, detector, visibility thresholds, and repeated-run protocol will be frozen before test evaluation.

A.8 REPRODUCIBILITY CHECKLIST

The completed release will report:

- source dataset versions, licenses, mappings, filters, and canonical-record statistics;
- HSSD and Habitat versions, scene identifiers, object assets, placement checks, and NavMesh settings;
- split manifests, paired-example identifiers, checksums, and leakage-audit output;
- model width, layers, attention heads, time features, history length, optimizer, learning rate, epochs, seeds, and checkpoint rule;
- detector and depth inputs, visibility computation, candidate and action budgets, evidence likelihoods, and planner hyperparameters;
- hardware, wall-clock latency, memory growth, repeated-run aggregation, and confidence intervals; and
- reproduced versus quoted baseline values, grouped results, failure labels, and all component-aligned ablations.

A.9 SUBMISSION COMPLIANCE NOTES

This source is configured for anonymous ICLR 2027 submission: \iclrfinalcopy remains disabled, author identities and affiliations are absent, and references precede the appendix. Before submission, authors should remove all result placeholders, keep the main text within the conference page limit, and scan figure metadata, linked repositories, PDF properties, acknowledgements, and supplementary files for identifying information.

A.10 ETHICS, REPRODUCIBILITY, AND AI USE

Ethics. Persistent visual memory may capture people or private spaces, while an incorrect belief may induce unnecessary motion or unsafe behavior. Deployments should obtain consent, enforce retention and access controls, preserve provenance, and retain platform-specific collision avoidance and emergency stops outside the learned model. Our benchmark compiles existing records into simulated HSSD episodes and introduces no new human-subject study.

Reproducibility. Section 3 specifies the task, memory, predictor, evidence update, and navigation policy. Section 4 defines episode construction, leakage controls, baselines, and metrics. An anonymized code and data snapshot will accompany the review submission.

AI use. Generative AI tools assisted manuscript editing, LaTeX checks, and local engineering. They did not generate benchmark observations, ground-truth labels, or reported scores. The authors reviewed all assisted material and remain responsible for the content and artifacts.